CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 9: N-Queens Problem using Backtracking

Course: Design Analysis and Algorithms Laboratory [23CS3505]
Category: Experiments / Practicals | Experiment #9 | Academic Year: 2026-27

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1. Overview & Objective:

Place N non-attacking chess queens on an Nˆ N board using bounding function pruning.

2. Practical Tasks & Verification Steps:

1. Represent board configuration using single 1D array board[N] where index is row and value is column.
2. Implement isSafe row, col function verifying column conflict and diagonal conflicts in O row time.
3. Write recursive solveNQueens row placing queen, advancing to next row, and backtracking on dead ends.
4. Print all distinct valid chessboard solutions for N = 4 and N = 8 boards.

Submission Guidelines & Verification:

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